

VMS Phase 1 Task Register — Business Partner & Vehicle

2026-09-20 · @Someone

How to read this register

136 tasks across 8 stages, estimated at **372 developer-days**. This covers everything specified in the VMS Phase 1 FSD for Business Partner and Vehicle — including the recurring charges, document management and role-based access sections — and nothing beyond it.

Convention	Meaning
Task ID	<code>S{stage}-{module}-{nn}</code> — S1-BP-03 is stage 1, Business Partner, task 3
Est	Developer-days for one developer, including unit tests and code review, excluding UAT
Depends	Task IDs that must complete first. Blank = can start once the stage opens
Spec	The FSD section or requirement ID this task implements
Layer	DB, API, UI, JOB (background), CFG (configuration/seed), QA

Effort by stage

Stage	Scope	Tasks	Est (days)
S0	Foundation — solution, database, auth, audit, numbering, storage, masters	18	46
S1	Business Partner — backend and frontend	24	62
S2	Vehicle — backend and creation wizard	26	74

Stage	Scope	Tasks	Est (days)
S3	Acquisition, finance agreement, opening postings	14	42
S4	Recurring charges and payables workbench	16	44
S5	Document management	15	40
S6	Role-based access control	12	32
S7	Notifications	6	14
S8	Testing, migration, UAT, go-live	5	18
	Total	136	372

With three developers working in parallel where dependencies allow, the critical path runs about 17 to 19 weeks. The dependency map in the last section shows where parallelism is possible and where it is not.

Estimates assume the team is already familiar with the stack and that the lookup masters are populated by the client, not researched by the team.

Stage 0 — Foundation

Nothing in stages 1 to 8 can start without this. It is the stage most often underestimated, because none of it is visible to the client.

ID	Task	Layer	Depends	Est	Spec
S0-FND-01	Solution scaffolding: project structure, DI, logging, configuration, error handling middleware	API		3	§24.3
S0-FND-02	Database project, connection management, migration tooling, seed runner	DB	S0-FND-01	2	§24.3
S0-FND-03	Authentication: login, password hashing, session or token issue, refresh, logout	API	S0-FND-01	4	§23B.1
S0-FND-04	User and Role tables, UserRole with branch scope	DB	S0-FND-02	2	§23B.1
S0-FND-05	Permission catalogue table and seeding of all ~55 permission codes	DB, CFG	S0-FND-04	2	§23B.2

ID	Task	Layer	Depends	Est	Spec
S0-FND-06	Authorisation middleware: endpoint permission attributes, default-deny, denial logging	API	S0-FND-05	4	BR-SEC-005, 006, 007
S0-FND-07	Field-level permission filter: server-side stripping of restricted fields from responses	API	S0-FND-06	3	BR-SEC-001, 002
S0-FND-08	Audit log table, append-only enforcement, change-capture interceptor with old/new values	DB, API	S0-FND-02	4	§13.2, BR-BP-022
S0-FND-09	Numbering series master and gap-free allocator inside the save transaction	DB, API	S0-FND-02	3	§24.1
S0-FND-10	File storage service: upload, virus scan hook, SHA-256 hash, generated naming, short-lived download URL	API	S0-FND-01	4	§23A.5
S0-FND-11	Date and time handling: UTC storage, client time zone rendering, date-only business dates, ISO 8601 API contract	API, UI	S0-FND-01	3	§24.4
S0-FND-12	Generic lookup master framework: table, CRUD API, admin screen, active flag and sort order	DB, API, UI	S0-FND-02	4	§24.2
S0-FND-13	Seed all lookup masters listed in §24.2	CFG	S0-FND-12	2	§24.2
S0-FND-14	City and Province master with mapping	DB, CFG	S0-FND-12	1	§24.2
S0-FND-15	Branch master and branch-scoped query filter	DB, API	S0-FND-12	2	§23B.4
S0-FND-16	Frontend shell: routing, layout, menu driven by permissions, auth guards	UI	S0-FND-06	4	§23B
S0-FND-17	Shared UI components: data grid with server paging, filter bar, lookup picker, file upload, date picker, confirm dialog	UI	S0-FND-16	5	§9.1, §21
S0-FND-18	Message resource file, validation display pattern, toast and error conventions	UI, API	S0-FND-17	2	§12.2

Stage 0 total: 18 tasks, 46 days.

S0-FND-07 is worth flagging: field-level stripping is the task that makes the profit-and-cost restrictions real rather than cosmetic. If it is skipped or deferred, every later screen has to re-solve it, and the exports will leak the amounts.

Stage 1A — Business Partner backend

ID	Task	Layer	Depends	Est	Spec
S1-BP-01	BusinessPartner table, BPCode series, Party Type, identity and tax fields, status, indexes	DB	S0-FND-09	3	§5, §6
S1-BP-02	BusinessPartnerRole table with effective dating and IsActive	DB	S1-BP-01	2	§5, BR-BP-001
S1-BP-03	Role detail tables: BPDriverDetail, BPVendorDetail, BPCustomerDetail	DB	S1-BP-02	2	§5, §7
S1-BP-04	Child tables: BPContact, BPAddress, BPBankAccount with IsPrimary constraints	DB	S1-BP-01	3	§8
S1-BP-05	Create partner API: single transaction across master, roles, details, children, opening balance, audit	API	S1-BP-04, S1-BP-03	5	§10, FR-BP-013
S1-BP-06	Update partner API with field-level change capture	API	S1-BP-05	3	§13.2
S1-BP-07	Core field validation: CNIC and NTN format, mobile format, conditional mandatories by role	API	S1-BP-05	3	§6, §12.2
S1-BP-08	Uniqueness enforcement for CNIC, NTN, STRN, licence number with friendly messages	API, DB	S1-BP-07	2	BR-BP-013
S1-BP-09	Duplicate detection service: exact hard checks plus name similarity scoring	API	S1-BP-08	4	§12.1
S1-BP-10	Soft-duplicate override capture and audit	API	S1-BP-09	1	BR-BP-021
S1-BP-11	Role add and remove with in-use guard and role change log	API	S1-BP-06	4	BR-BP-010, 011, 016
S1-BP-12	Status change API: deactivate, reactivate, blacklist with reason and open-record guard	API	S1-BP-06	3	§13.1, BR-BP-014
S1-BP-13	Opening balance posting at creation, field lock after save	API	S1-BP-05	2	BR-BP-018
S1-BP-14	Partner list API: search across code, name, CNIC, NTN, mobile; filters; server paging and sorting	API	S1-BP-05	4	§9.1

ID	Task	Layer	Depends	Est	Spec
S1-BP-15	Role-filtered picker API used by every partner dropdown elsewhere	API	S1-BP-02	2	FR-BP-003, BR-BP-020
S1-BP-16	Linked Vehicles read API for the partner screen	API	S2-VH-03	2	§9.2
S1-BP-17	Partner export to Excel honouring field permissions	API	S1-BP-14, S0-FND-07	2	BR-SEC-003

Stage 1A total: 17 tasks, 47 days.

S1-BP-16 depends on a stage 2 table, so it is scheduled last in this stage or moved into stage 2. Everything else in 1A is self-contained once stage 0 lands.

Stage 1B — Business Partner frontend

ID	Task	Layer	Depends	Est	Spec
S1-BPU-01	Partner list screen: columns, quick search, filters, row actions, empty state	UI	S1-BP-14, S0-FND-17	4	§9.1
S1-BPU-02	Partner form shell: tab layout, header with code, name, role chips and status, dirty-state guard	UI	S0-FND-17	3	§9.2, FR-BP-011
S1-BPU-03	General tab: core fields, Party Type switching CNIC and NTN, live mandatory indicators	UI	S1-BPU-02	4	§6, FR-BP-009
S1-BPU-04	Role multi-select driving panel visibility without reload	UI	S1-BPU-03	2	FR-BP-008
S1-BPU-05	Role panels: Driver, Workshop, Bank, Vendor, Customer, Running Customer, Tracker, Body Maker, Fuel Card	UI	S1-BPU-04	6	§7
S1-BPU-06	Duplicate warning panel with candidate comparison and override tick	UI	S1-BP-09	3	§12.1
S1-BPU-07	Contacts and Addresses grids with inline add, edit and primary toggle	UI	S1-BPU-02	3	§8.1, §8.2
S1-BPU-08	Bank Accounts grid with IBAN validation	UI	S1-BPU-02	2	§8.3
S1-BPU-09	Client-side holding of child rows until first save, then single-transaction submit	UI	S1-BPU-07	3	FR-BP-010

ID	Task	Layer	Depends	Est	Spec
S1-BPU-10	Compact partner dialog for inline creation from any picker, with role pre-selected and locked	UI	S1-BPU-03	3	FR-BP-014, §10
S1-BPU-11	Status change dialog with reason capture and blocking-record list	UI	S1-BP-12	2	§13.1
S1-BPU-12	History tab: audit entries, role change log, status change log	UI	S0-FND-08	3	§9.2
S1-BPU-13	Linked Vehicles tab	UI	S1-BP-16	2	§9.2
S1-BPU-14	Save and New behaviour retaining Party Type, roles, city and branch	UI	S1-BPU-09	1	FR-BP-015

Stage 1B total: 14 tasks, 41 days.

S1-BPU-05 is the largest single UI task in the register: nine role panels, each with its own conditional mandatories. It is worth splitting across two developers by role group.

Merge Partners (FR-BP-017) is deliberately **not** in this stage. It is listed under stage 8 as deferrable pending OQ-06.

Stage 2A — Vehicle backend

ID	Task	Layer	Depends	Est	Spec
S2-VH-01	Vehicle table: identity, technical, operational fields, status, current category copies, indexes	DB	S0-FND-09	4	§15, §16
S2-VH-02	Registration number normalisation and filtered unique index across non-disposed vehicles	DB, API	S2-VH-01	2	BR-VH-017
S2-VH-03	VehicleBusinessPartnerRelation with filtered unique index on one open relation	DB	S2-VH-01, S1-BP-01	3	BR-VH-002
S2-VH-04	VehicleLifecycleHistory table and status transition service	DB, API	S2-VH-01	3	§23.1
S2-VH-05	VehicleAttachedItem table with attach, detach and transfer chain	DB, API	S2-VH-01	4	§20.1, BR-VH-013
S2-VH-06	OdometerReading table with floor validation against opening reading	DB, API	S2-VH-01	2	BR-VH-025

ID	Task	Layer	Depends	Est	Spec
S2-VH-07	DriverVehicleAssignment with single-active-vehicle guard and release prompt	DB, API	S2-VH-01, S1-BP-03	3	BR-VH-022
S2-VH-08	Save Draft API: vehicle row only, no postings	API	S2-VH-01	2	FR-VH-012, BR-VH-012
S2-VH-09	Activate API: atomic write across vehicle, relation, lifecycle, postings, items, documents	API	S2-VH-03, S3-FIN-06	6	FR-VH-012, §19
S2-VH-10	Activation pre-check service returning the checklist of what is missing	API	S2-VH-09	3	BR-VH-019
S2-VH-11	Category-specific validation for Shared, Rented, Customer Arrangement	API	S2-VH-03	4	§17.1-17.3
S2-VH-12	Change Category action: close relation, open new, lifecycle row, guards on open finance	API	S2-VH-03	4	BR-VH-005, 006, 007
S2-VH-13	Dispose actions: retire, sell, transfer with counterparty, amount and reason	API	S2-VH-04	3	BR-VH-023
S2-VH-14	Vehicle list API: columns, filters, quick search across reg, chassis, engine, code	API	S2-VH-01	4	§21
S2-VH-15	Vehicle financial summary API computing paid-to-date and outstanding as aggregates	API	S3-FIN-07	3	BR-VH-003, §19
S2-VH-16	Fuel card uniqueness and reassignment prompt	API	S2-VH-01	2	BR-VH-026

Stage 2A total: 16 tasks, 52 days.

S2-VH-09 is the highest-risk task in the register. It writes to eight tables in one transaction and must roll back completely on any failure. Budget the full six days and write the failure tests first.

Stage 2B — Vehicle frontend

ID	Task	Layer	Depends	Est	Spec
S2-VHU-01	Wizard framework: five steps, progress bar, per-step error counts, step gating	UI	S0-FND-17	4	§21, FR-VH-009
S2-VHU-02	Step 1 Vehicle details: identity and technical fields, conditional capacity mandates	UI	S2-VHU-01	4	§16.1, §16.2

ID	Task	Layer	Depends	Est	Spec
S2-VHU-03	Copy from existing vehicle pre-fill	UI	S2-VHU-02	2	FR-VH-014
S2-VHU-04	Step 2 Ownership: category selector driving five conditional field sets	UI	S2-VHU-01, S2-VH-11	5	§17
S2-VHU-05	Counterparty picker filtered by role, with inline partner creation	UI	S2-VHU-04, S1-BPU-10	2	FR-VH-013, BR-VH-020
S2-VHU-06	Step 4 Items and documents grids	UI	S2-VHU-01	4	§20
S2-VHU-07	Step 5 Review: read-only summary plus the list of postings about to be written	UI	S2-VHU-01, S3-FIN-07	4	FR-VH-011
S2-VHU-08	Save Draft and Activate actions with the missing-items checklist	UI	S2-VH-10	3	FR-VH-012, BR-VH-019
S2-VHU-09	Vehicle list screen: columns, filters, quick search, row actions	UI	S2-VH-14	4	§21
S2-VHU-10	Vehicle screen shell: header, tab layout, financial summary panel	UI	S2-VH-15	5	§5 of source, §19
S2-VHU-11	Change Category dialog with dated close and open	UI	S2-VH-12	3	BR-VH-005
S2-VHU-12	Attached Items tab with attach, detach and transfer	UI	S2-VH-05	3	§20.1
S2-VHU-13	Driver assignment with release prompt	UI	S2-VH-07	2	BR-VH-022
S2-VHU-14	Dispose dialogs: retire, sell, transfer	UI	S2-VH-13	3	BR-VH-023
S2-VHU-15	Vehicle History tab replaying the induction as one dated group	UI	S0-FND-08	3	§23.2
S2-VHU-16	Vehicle export honouring field permissions	UI, API	S2-VH-14, S0-FND-07	2	BR-SEC-003

Stage 2B total: 16 tasks, 53 days.

S2-VHU-04 carries the conditional complexity of the whole module — five categories, each with a different field set and its own validation. It is the task most likely to need a second pass after the client sees it.

Stage 3 — Acquisition, finance and opening postings

This stage delivers the §8 worked example: PKR 2,000,000 at creation plus a PKR 150,000 installment showing PKR 2,150,000 paid, with no stored total anywhere.

ID	Task	Layer	Depends	Est	Spec
S3-FIN-01	VehicleTransaction table: type, amount, date, partner, reference, source, IsSystemGenerated, attachment	DB	S2-VH-01	3	§6 of source, §19
S3-FIN-02	VehicleFinanceAgreement table with terms, one-active-per-vehicle guard	DB	S2-VH-01, S1-BP-01	3	§18.2, BR-VH-004
S3-FIN-03	VehicleInstallment table with status and paid tracking	DB	S3-FIN-02	2	§18.3
S3-FIN-04	Acquisition block API and validation, including amount-paid ceiling	API	S3-FIN-01	3	§18.1, BR-VH-021
S3-FIN-05	Finance block API with down-payment equality check and reconciliation warning	API	S3-FIN-02	3	BR-VH-009, BR-VH-010
S3-FIN-06	Installment schedule generator: frequency stepping, month-end fallback, residual row	API	S3-FIN-03	4	FR-VH-004
S3-FIN-07	Opening posting engine: acquisition, initial payment, registration cost, deposit, item costs	API	S3-FIN-04, S3-FIN-06	5	§19
S3-FIN-08	Derived totals service: paid-to-date, outstanding, total payable	API	S3-FIN-07	3	FR-VH-006, BR-VH-003
S3-FIN-09	Record installment payment API updating schedule and posting a transaction	API	S3-FIN-03	3	§8 of source
S3-FIN-10	Adjustment and reversal route with reason, audit and no physical delete	API	S3-FIN-07	4	BR-VH-011, source §13
S3-FIN-11	Step 3 UI: acquisition fields, finance block, live total payable	UI	S3-FIN-05	4	§18
S3-FIN-12	Schedule preview grid with editable due dates before save	UI	S3-FIN-06	3	FR-VH-005
S3-FIN-13	Vehicle Finance tab: agreement summary, schedule, payment history, outstanding	UI	S3-FIN-08	4	§5 of source
S3-FIN-14	Record installment payment dialog with receipt upload	UI	S3-FIN-09	2	§7 of source

Stage 3 total: 14 tasks, 46 days.

S3-FIN-08 should be covered by tests written directly from AC-VH-003, AC-VH-004 and AC-VH-005. Those three acceptance criteria are the client's stated requirement, and if the derived totals pass them the module's financial behaviour is demonstrably correct.

Stage 4 — Recurring charges

ID	Task	Layer	Depends	Est	Spec
S4-REC-01	VehicleRecurringCharge table with effective dating for amount changes	DB	S2-VH-01	3	§19A.1, BR-VH-033
S4-REC-02	VehicleRecurringChargeEntry table with period key and unique constraint	DB	S4-REC-01	2	BR-VH-030
S4-REC-03	Charge configuration API with payee role filtering and posting-mode rules	API	S4-REC-01	4	§19A.1, BR-VH-027
S4-REC-04	Auto-post permission gate and fixed-amount-only restriction	API	S4-REC-03, S0-FND-06	2	BR-VH-027, BR-VH-028
S4-REC-05	Nightly generation job: idempotent on charge and period, lead-day windowing	JOB	S4-REC-02	5	BR-VH-030
S4-REC-06	Bank installment surfacing: read VehicleInstallment as Due, no parallel schedule	JOB, API	S4-REC-05, S3-FIN-03	3	BR-VH-029
S4-REC-07	Duplicate-charge guard blocking a manual Bank Installment charge on a financed vehicle	API	S4-REC-06	1	VAL-VH-019
S4-REC-08	Confirm payment API: post one transaction, retain expected vs actual, link back	API	S4-REC-02, S3-FIN-01	4	BR-VH-031
S4-REC-09	Waive and cancel actions with reason and audit	API	S4-REC-08	2	§19A.4
S4-REC-10	Overdue transition job and escalation flagging	JOB	S4-REC-05	2	§19A.4
S4-REC-11	End-dating cascade on vehicle disposal and category change	API	S4-REC-01, S2-VH-12	3	BR-VH-034, BR-VH-035
S4-REC-12	Recurring Charges grid in wizard step 4 and on the vehicle screen	UI	S4-REC-03	4	§19A.5
S4-REC-13	Due and Payments panel on the vehicle screen	UI	S4-REC-08	3	§19A.5
S4-REC-14	Payables Due workbench: fleet-wide list, filters, sorting	UI	S4-REC-08	4	FR-VH-016

ID	Task	Layer	Depends	Est	Spec
S4-REC-15	Bulk confirm with common payment date and mode	UI, API	S4-REC-14	3	§19A.5
S4-REC-16	Dashboard tile: due within 7 days, overdue count and total	UI	S4-REC-14	2	§19A.5

Stage 4 total: 16 tasks, 47 days.

S4-REC-05 and S4-REC-06 together are the subtle part. The generation job must treat the lease schedule as a source it reads rather than a schedule it owns, or the fleet ends up with two competing installment records per month.

Stage 5 — Document management

ID	Task	Layer	Depends	Est	Spec
S5-DOC-01	DocumentType master: applies-to, expirable, validity, periodic, lead days, mandatory, retention	DB, CFG	S0-FND-12	3	§23A.1
S5-DOC-02	Seed the twelve document types with their periodicity settings	CFG	S5-DOC-01	1	§23A.2
S5-DOC-03	Document table serving both owners, with version number, IsCurrent, supersedes link, status	DB	S5-DOC-01, S0-FND-10	4	§23A.3
S5-DOC-04	Upload API: type-driven validation, format and size checks, hash, scan, audit	API	S5-DOC-03	4	§23A.5, BR-DOC-007
S5-DOC-05	Renewal API: create version n+1, flip IsCurrent in one transaction	API	S5-DOC-04	3	BR-DOC-001
S5-DOC-06	Expiry computation job: nightly status recalculation to Active, Expiring Soon, Expired	JOB	S5-DOC-03	2	BR-DOC-003
S5-DOC-07	Mandatory-document check feeding vehicle activation and partner save	API	S5-DOC-03, S2-VH-10	2	BR-VH-015, BR-BP-005
S5-DOC-08	Reject action with reason, excluded from mandatory satisfaction	API	S5-DOC-04	2	BR-DOC-008
S5-DOC-09	Retention job removing superseded versions past their type's retention, with a log	JOB	S5-DOC-03	2	BR-DOC-002

ID	Task	Layer	Depends	Est	Spec
S5-DOC-10	Download permission split and audited download endpoint	API	S5-DOC-04, S0-FND-06	2	§23A.5
S5-DOC-11	Documents tab for partner and vehicle: current by type, status chips, history toggle	UI	S5-DOC-04	5	§23A.4
S5-DOC-12	Renew dialog with pre-filled expiry and optional expense posting linked to a recurring charge	UI	S5-DOC-05, S4-REC-08	4	BR-DOC-005, BR-DOC-006
S5-DOC-13	Document Register screen with full filter set	UI	S5-DOC-06	4	§23A.4
S5-DOC-14	Missing Documents report	UI, API	S5-DOC-07	3	§23A.4
S5-DOC-15	Expiry Calendar month view	UI	S5-DOC-06	3	§23A.4

Stage 5 total: 15 tasks, 44 days.

Bulk upload is intentionally excluded from this stage and sits in stage 8 with the migration work, since its only real purpose is go-live loading.

S5-DOC-12 is where document management meets the payables engine. Get the link right and an insurance renewal is recorded once; get it wrong and the premium appears twice in the vehicle's P&L.

Stage 6 — Role-based access control

The enforcement primitives are built in stage 0 (S0-FND-05 to S0-FND-07). This stage delivers the configuration surface and applies the permissions across every screen already built.

ID	Task	Layer	Depends	Est	Spec
S6-SEC-01	Role CRUD API with system-role protection and last-admin guard	API	S0-FND-04	3	BR-SEC-008
S6-SEC-02	RolePermission assignment API with audit of every grant and revoke	API	S6-SEC-01	3	BR-SEC-010
S6-SEC-03	User CRUD, role assignment, branch scope, link to a Driver partner	API	S6-SEC-01, S1-BP-03	4	§23B.1

ID	Task	Layer	Depends	Est	Spec
S6-SEC-04	Effective permission resolution: union across roles, session re-evaluation on change	API	S6-SEC-02	3	BR-SEC-009
S6-SEC-05	Data scope filters applied to partner, vehicle and transaction queries	API	S0-FND-15, S6-SEC-03	4	§23B.4
S6-SEC-06	Apply field permissions to every existing response: cost, finance, profit, salary, credit	API	S0-FND-07, S3-FIN-08	4	BR-SEC-001, 002
S6-SEC-07	Derived-value suppression: hide totals, charts and exports that would reveal a hidden field	API	S6-SEC-06	3	BR-SEC-002
S6-SEC-08	Separation-of-duties switch for approvals and adjustments	API	S6-SEC-04	2	BR-SEC-011
S6-SEC-09	Role management screen with cloning	UI	S6-SEC-01	3	FR-SEC-001
S6-SEC-10	Permission tree editor grouped by module and level, with search and select-all	UI	S6-SEC-02	4	§23B.6
S6-SEC-11	User management screen with role and scope assignment	UI	S6-SEC-03	3	§23B.6
S6-SEC-12	Effective permissions viewer showing which role granted each capability	UI	S6-SEC-04	3	§23B.6
S6-SEC-13	Seed the six default role templates	CFG	S6-SEC-02	2	§23B.7

Stage 6 total: 13 tasks, 41 days.

S6-SEC-06 and S6-SEC-07 touch code written in stages 1 to 5, which is why this stage sits late. The alternative — applying field permissions as each screen is built — spreads the same work across earlier stages but makes each of them harder to test in isolation. Either sequencing works; what does not work is leaving it until after UAT.

S6-SEC-12 pays for itself the first time someone asks why a user can see a figure they should not.

Stage 7 — Notifications

Delivery channels beyond in-app are a separate FSD. This stage builds the rule engine and the subscriptions that the two modules create, so nothing specified here is left dangling.

ID	Task	Layer	Depends	Est	Spec
S7-NOT-01	NotificationRule master: event type, lead days, recipients, escalation, active flag	DB, CFG	S0-FND-12	3	§9 of source
S7-NOT-02	Subscription creation at save time for document expiry, installment due and charge due	API	S5-DOC-04, S4-REC-05	3	FR-BP-016, FR-VH-008
S7-NOT-03	Nightly evaluation job producing notifications against the configured lead times	JOB	S7-NOT-01	3	§23.4, §19A.6
S7-NOT-04	Operating time zone setting for the batch, so a lead time means the same day for every user	CFG, JOB	S7-NOT-03	1	NFR-DT-06
S7-NOT-05	Notification rule admin screen	UI	S7-NOT-01	3	§9 of source
S7-NOT-06	In-app notification list and unread indicator	UI	S7-NOT-03	3	§23.4

Stage 7 total: 6 tasks, 16 days.

S7-NOT-04 is one day of work that prevents a class of confusing bugs. Without it the reminder date drifts by user location.

Stage 8 — Testing, migration and go-live

ID	Task	Layer	Depends	Est	Spec
S8-QA-01	Automated acceptance suite covering all 24 criteria in §25 of the FSD	QA	S6-SEC-07	6	§25
S8-QA-02	Migration tooling for existing partner and vehicle records, with a dry-run and reconciliation report	API, QA	S2-VH-09	5	OQ-09
S8-QA-03	Bulk document upload screen for go-live loading	UI	S5-DOC-04	3	§23A.4
S8-QA-04	UAT support, defect triage and fixes	QA	S8-QA-01	6	—
S8-QA-05	Go-live: environment setup, master data load, user and role creation, cutover	CFG	S8-QA-04	3	—

Stage 8 total: 5 tasks, 23 days.

Deferred, pending client answers

These are specified in the FSD but deliberately excluded from the 372-day estimate. Each becomes a task only if the client confirms it for V1.

ID	Task	Est	Gate
D-01	Merge Partners: re-point children and transactions, tombstone the duplicate	6	OQ-06 — if V1 rather than Phase 1.1
D-02	Partner import from Excel	4	Client request, currently Phase 1.1
D-03	Maker-checker workflow on expenses and adjustments	5	OQ-19
D-04	Principal and markup split on lease installments	4	OQ-07
D-05	Driver salary as a vehicle-level charge with allocation	3	OQ-14

If all five are confirmed, add 22 days and roughly two weeks to the critical path.

Dependencies and critical path

flowchart LR

```

S0[S0 Foundation<br/>46d] --> S1[S1 Business Partner<br/>88d]
S0 --> S2A[S2A Vehicle backend<br/>52d]
S1 --> S2A
S2A --> S3[S3 Finance<br/>46d]
S3 --> S2B[S2B Vehicle wizard<br/>53d]
S3 --> S4[S4 Recurring<br/>47d]
S0 --> S5[S5 Documents<br/>44d]
S4 --> S5
S2B --> S6[S6 RBAC<br/>41d]
S5 --> S6
S4 --> S7[S7 Notifications<br/>16d]
S6 --> S8[S8 QA and go-live<br/>23d]
S7 --> S8

```

The critical path

S0 → S2A → S3 → S2B → S6 → S8. Everything on it is sequential because each stage consumes something the previous one creates: the vehicle wizard cannot show its posting preview until the posting engine exists, and the field-permission pass cannot run until the screens it hides fields on are built.

Position	Stage	Why it cannot move earlier
1	S0 Foundation	Auth, audit, numbering and storage underpin everything
2	S2A Vehicle backend	Needs the partner tables for counterparty relations
3	S3 Finance	Vehicle activation is incomplete without the posting engine
4	S2B Vehicle wizard	Step 5's posting preview needs S3-FIN-07
5	S6 RBAC	Applies field permissions to screens from S1 to S5
6	S8 QA and go-live	Acceptance tests need the permission behaviour in place

What can run in parallel

- **S1B (partner frontend)** runs alongside S2A once S1A's APIs are stable. Different developer, no contention.
- **S5 (documents)** needs only stage 0 plus the recurring-charge link in S5-DOC-12. Start it during S3 and hold that one task.
- **S7 (notifications)** is independent once S4 and S5 have created their subscriptions.
- **S8-QA-02 (migration)** can begin as soon as S2-VH-09 exists, well before UAT.

With one developer the register runs about 74 weeks. With three it compresses to roughly 17 to 19 weeks, limited by the critical path rather than by available hands. A fourth developer adds little.

Milestones

Six demonstrable points. Each is chosen because the client can see something work, not because a stage boundary was reached.

#	Milestone	Completes	What the client can do
M1	Platform ready	S0	Log in, see a permission-driven menu, manage lookup masters. Nothing business-facing yet
M2	Partners live	S1A, S1B	Create every partner type with multi-role, contacts, addresses, documents. Duplicate detection works. Data entry for go-live can begin here, in parallel with the rest of the build
M3	Vehicles inducted	S2A, S2B, S3	Run the full five-step wizard, including a bank lease with its generated schedule, and see the §8 worked example produce PKR 2,150,000 after one installment
M4	Payables running	S4	See this month's due items across the fleet on one workbench and clear them in bulk
M5	Compliance visible	S5, S7	The document register shows what expires when, and reminders fire on the configured lead times
M6	Controlled and accepted	S6, S8	Each role sees only its permitted screens, operations and amounts; all 24 acceptance criteria pass

M2 is the one to aim for early. Partner data entry is the client's largest manual effort at go-live, and it can start the moment M2 lands rather than waiting for the whole system. Pulling S1 forward converts weeks of client data entry from sequential to parallel.

M3 is the client's confidence point. The §8 worked example is the requirement they wrote themselves. Demonstrating it end to end settles whether the financial model is right before four more stages are built on top of it.

Risks and blocked tasks

Tasks blocked by open questions

These cannot be finished until the client answers. Most can be started; none can be signed off.

Open question	Blocks	Impact if unanswered by
OQ-02 opening balances	S1-BP-13	Start of S1A
OQ-03 role-specific fields for Bank, Workshop, Body Maker, Tracker, Fuel Card	S1-BPU-05	Start of S1B
OQ-04 licence scan blocking or warning	S5-DOC-07	Start of S5
OQ-05 shared-vehicle expense split timing	S2-VHU-04	Start of S2B
OQ-07 principal and markup split	S3-FIN-06	Start of S3 — changes the schedule table
OQ-08 down payment treatment for P&L	S3-FIN-07	Start of S3 — changes the posting engine
OQ-09 records to migrate and in what format	S8-QA-02	4 weeks before go-live
OQ-10 branches and default branch restriction	S0-FND-15	Start of S0
OQ-12 auto-post defaults	S4-REC-04	Start of S4
OQ-15 expired document blocking trip assignment	S5-DOC-07	Start of S5
OQ-19 maker-checker	S6-SEC-08	Start of S6
OQ-20 driver app users in the same role system	S6-SEC-03	Start of S6

OQ-07 and OQ-08 are the urgent pair. Both change the shape of stage 3's tables and posting logic, and stage 3 sits on the critical path. Answering them late means reworking the posting engine after the wizard has been built against it.

OQ-10 is needed before stage 0 finishes, since branch scoping is wired into the query layer rather than bolted on.

Delivery risks

Risk	Likelihood	Effect	Response
S2-VH-09 atomic activation proves harder than estimated	Medium	Slips the critical path directly	Write the rollback failure tests before the implementation; budget the full 6 days
Lookup masters not populated by the client in time	High	Blocks every screen's testing, not just seeding	Request the lists at M1, not at go-live; ship with the seeds in this register as defaults
Field-permission pass (S6-SEC-06, 07) uncovers leaks in earlier screens	Medium	Rework across stages 1 to 5	Apply field permissions to each screen as it is built, rather than only in stage 6
Client requests category or role changes after seeing M3	Medium	S2-VHU-04 rework	Demonstrate the category screens at M3 specifically to surface this early
Recurring-charge and lease schedule produce duplicate due items	Low	Wrong payables figures, hard to spot	S4-REC-06 and S4-REC-07 exist for this; test with a financed vehicle carrying insurance and tracker charges
Estimates assume familiarity with the stack	Medium	Uniform overrun	Re-baseline after M1, when the first 46 days of actuals are known

What to do first

1. Send OQ-01 to OQ-20 to the client, flagging OQ-07, OQ-08 and OQ-10 as needed within the week.
2. Request the lookup master lists in §24.2 of the FSD, and the branch list.
3. Start S0-FND-01 to S0-FND-11 — none of them depend on any open question except OQ-10.